

management fee. To achieve the highest returns possible, managers need to understand the cost structure of each asset, and choose the most cost effective path to gain the desired exposure.

Example:

A portfolio manager looking to gain exposure to the general market index can purchase either stocks, exchange traded funds, or futures. How should the manager determine the best approach to acquire the exposure?

Based on what we have observed in practice, stocks often have the lowest cost for smaller sizes, followed by exchange traded funds for slightly larger sizes, and then futures for the largest sizes. This is illustrated in [Figure 12.7a](#). In this example, stocks are the most inexpensive option for investment dollars up to \$250 million. Exchange traded funds are the most inexpensive investment vehicle for investment dollars from \$250 million to \$667 million. And futures are the most inexpensive for investments of \$667 million and higher.

Note: These values and breakpoints are for illustration purposes only based on data from 2011. Investors need to apply techniques provided in this chapter to determine the exact break points for today's market conditions.

What causes these differences across investment vehicles?

Equities: Equities often have the lowest implementation cost for smaller dollar values but these costs increased at the fastest rates. Equity trading costs tend to increase at the fastest rates because buying or selling pressure often causes market participants to believe that the excessive transaction pressure is due to a mispricing of the stock price or due to changing company fundamentals that have not yet been fully disseminated into the market. The corresponding price change then often attracts momentum players and active managers hoping to achieve a short-term trading profit which further impacts the stock price.

Exchange Traded Funds: ETFs tend to have initial trading costs (intercept term) higher than the underlying stocks. This is primarily due to the trading cost corresponding with acquiring the position (similar to stocks) and also the management fee charged by the ETF fund manager for maintaining the appropriate ETF portfolio. ETF trading costs, however, increase at a slower rate with investment value than the underlying stocks due to shadow liquidity. Shadow liquidity refers to the potential trading volumes from market participants who stand ready to buy or sell shares in an ETF if there is a mispricing in the market. As soon as the ETF price